

# **MODULE I**

## **Engineering and Engineer**

# Content

- Engineering as a discipline and a profession
- Attributes and functions of a practicing engineer
- Engineer as a problem solver
- Engineer as a designer
- Engineer as a change agent

# Engineering as a Discipline and a Profession

# Introduction

- **Definition of Engineering**

Engineering is a professional discipline that combines knowledge from **physics, mathematics, and material science** with **innovation, analysis, and design** to develop **technologies and solutions** that meet **human needs** in a safe, efficient, and sustainable way.

- **Brief Overview:**

- Engineering as a body of knowledge (discipline)
- Engineering as a career path (profession)

## 7 most essential **skills** that every engineer needs to have



# WHAT DO ENGINEERS DO?

**ASK QUESTIONS**  
to define a problem or challenge

**BUILD** models and prototypes

Use digital tools and mathematical concepts to **TEST AND COMPARE SOLUTIONS**

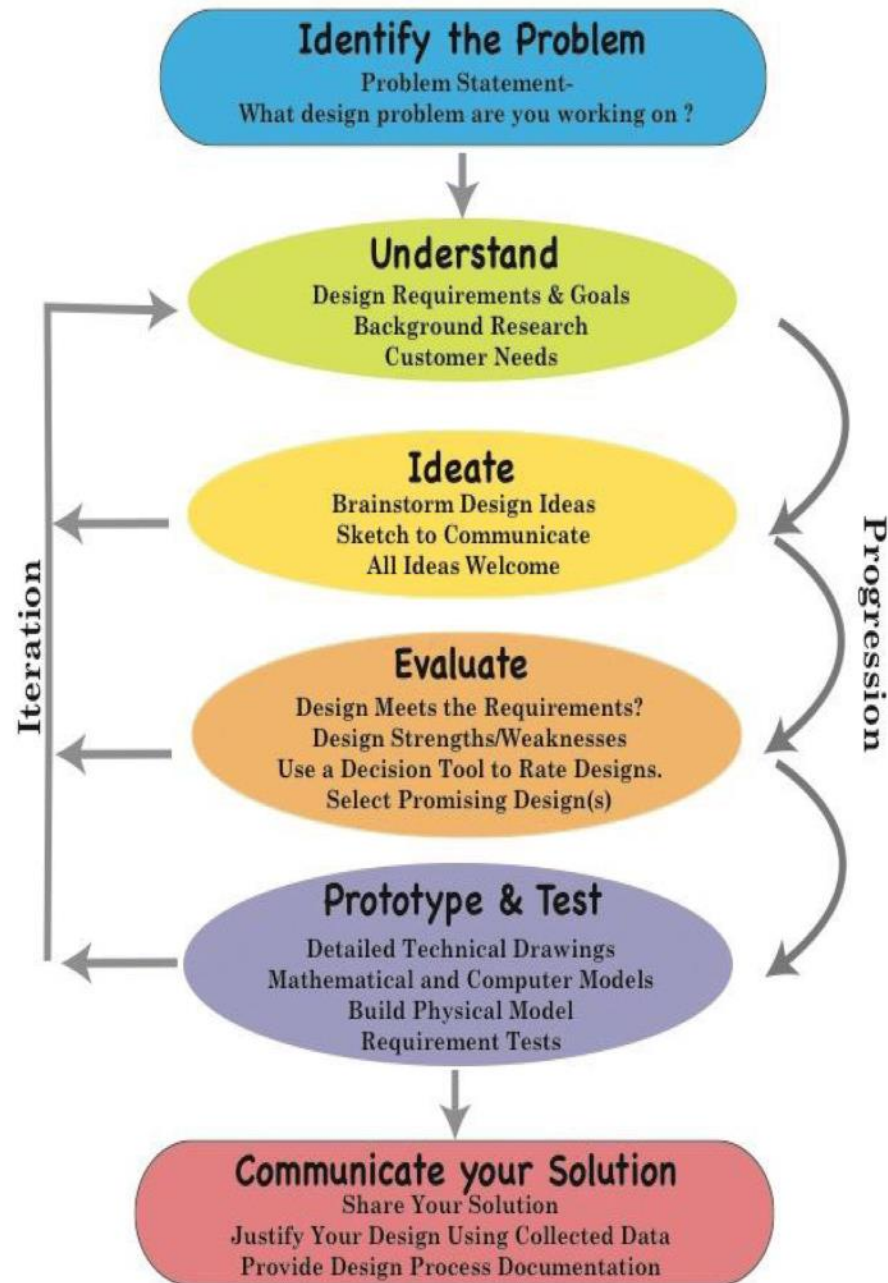
**ANALYZE DATA AND INTERPRET** results to identify points for improvement

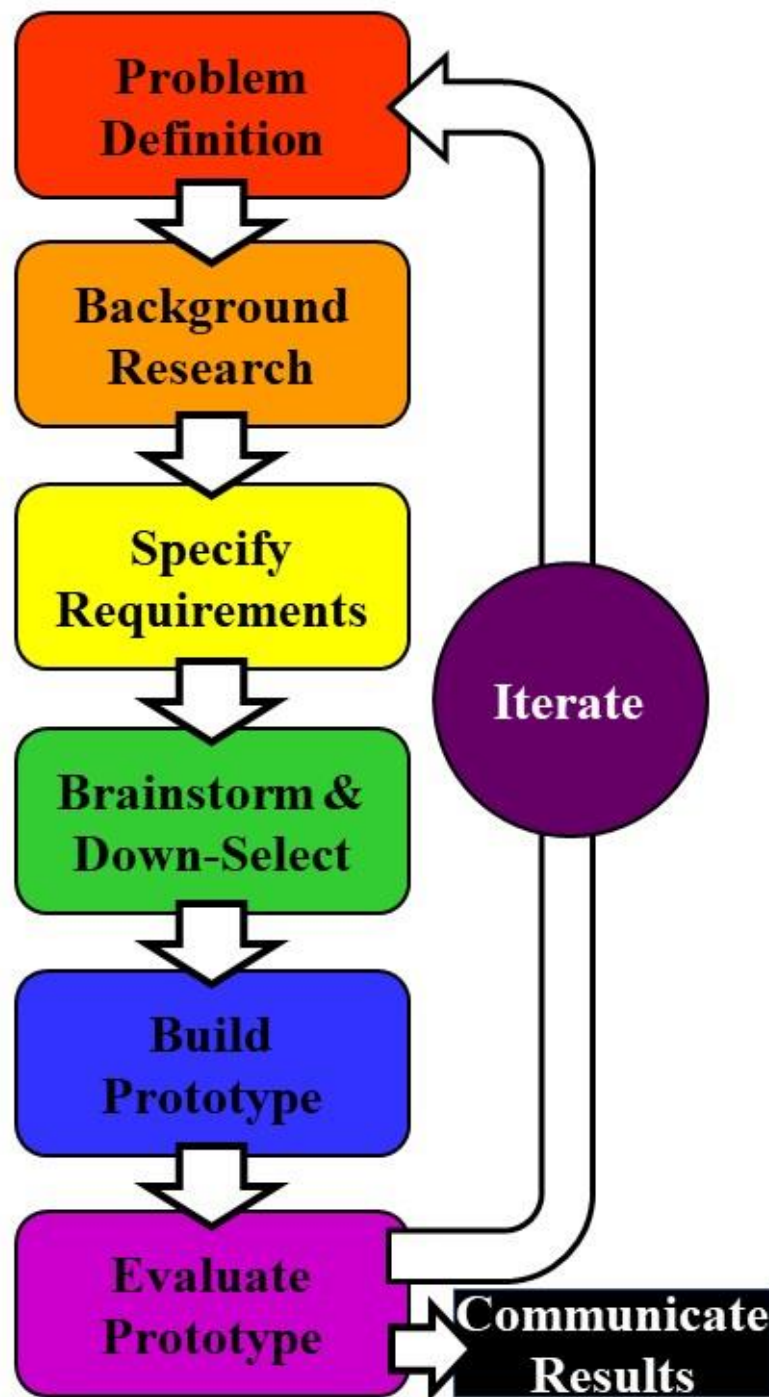
Design, evaluate, and **REFINE** solutions

Defend designs with evidence and **REVISE** when necessary

**COMMUNICATE** and compare design ideas and solutions

# Engineering Design Process







## **Theory**

### **COMPUTER SCIENCE**

- Programming languages
- Coding, cryptography
- Algorithms & numerical analysis
- Compilers
- Network security/privacy
- Theory of computation

## **Software**

- Operating systems
- Basic Computer architecture
- Low-level programming

## **Hardware**

### **COMPUTER ENGINEERING**

- Microprocessor/microcontroller systems
- Embedded digital systems
- Processor interfacing
- Wireless networking
- Digital logic design
- Solid state physics

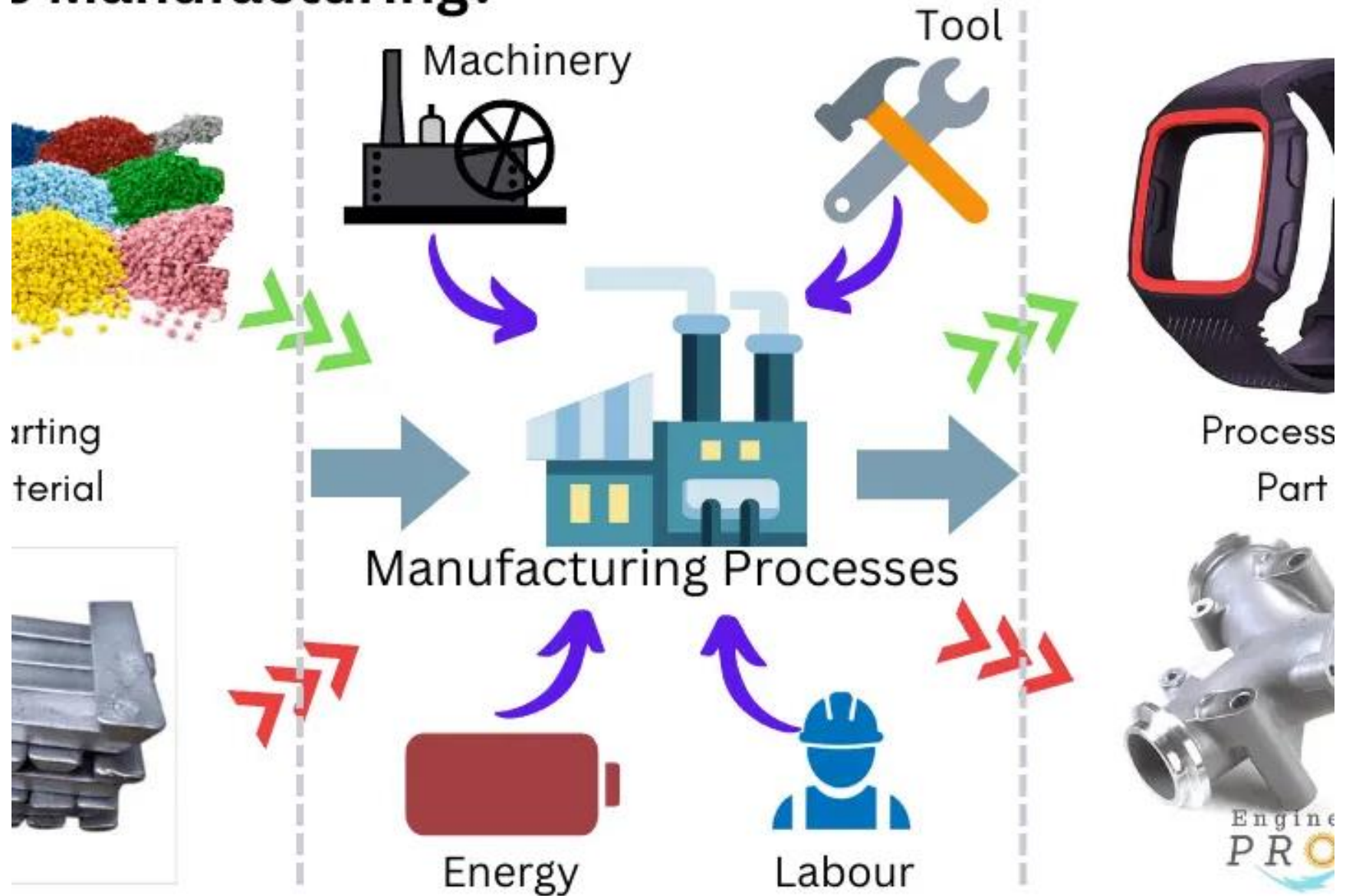
## Commonly-Used Methodology for Problem-Solving in Engineering



# Petrochemical



# What is Manufacturing?



# Engineering as a Discipline

- Engineering as a **discipline** refers to a **branch of academic study and systematic knowledge** that applies scientific and mathematical principles to solve real-world problems.
- Focus on problem-solving and innovation
- It involves learning theories, developing technologies, designing systems, and innovating solutions to improve life.
- **Key branches:**  
Mechanical, Electrical, Civil, Chemical, Computer, etc.





# Characteristics of Engineering Discipline

- Interdisciplinary nature
- Strong foundation in math and science.
- Involves design, analysis, experimentation.
- Emphasis on continual improvement and lifelong learning.

# Key Components of Engineering as a Discipline

## **1. Scientific Foundation:**

Engineers must understand the laws of nature—how forces act, how energy flows, and how materials behave. This knowledge comes from physics, chemistry, and biology.

## **2. Mathematical Modeling:**

Mathematics is the language of engineering. Engineers use equations and formulas to model systems, predict outcomes, and optimize performance.

### **3. Design Thinking:**

Engineering students are trained to think creatively and logically. They learn how to define a problem, brainstorm solutions, model and simulate designs, and improve systems through iteration.

### **4. Technical Knowledge:**

Each branch of engineering whether it's mechanical, electrical, civil, electronics, chemical, or computer has its own technical core. Students dive deep into specialized subjects relevant to that field.



## **5. Experimentation and Analysis:**

Labs and experiments help engineers validate theories. They collect data, analyze results, and understand how things work in real-world scenarios.

## **6. Innovation and Research:**

Engineering as a discipline encourages innovation developing new materials, creating energy efficient systems, improving manufacturing processes, and advancing technology through research.

# Examples in Academic Context:

- A **mechanical engineering student** learns about thermodynamics, machine design, and manufacturing processes.
- An **electrical engineering student** studies circuit theory, electromagnetism, and signal processing.
- A **computer science student** explores algorithms, programming languages, artificial intelligence, and data structures.

# Engineering as a Profession

- Engineering as a **profession** goes beyond classroom learning—it is the **practical application** of engineering knowledge to **design, develop, operate, and maintain** systems, products, and processes that serve the needs of society.
- A professional engineer takes on real-world challenges and works to solve them responsibly, ethically, and efficiently.
- Application of engineering knowledge for societal benefit.
- Licensed and regulated by professional bodies (e.g., IEEE, ASCE, NEC).
- Career-oriented focus: roles in industry, government, and consultancy.

# Key Elements of Engineering as a Profession

## 1. Application of Knowledge:

Engineers use the theories and principles learned during their academic training to solve problems in industries such as construction, manufacturing, energy, healthcare, information technology, and transportation.

Example: A mechanical engineer designs an energy-efficient HVAC system for a hospital using thermodynamics and fluid flow principles.

## 2. Professional Responsibility:

Engineering is not just about solving technical problems; it also **involves making decisions that affect people, property, and the environment**. Engineers are trusted to act in the public interest.

Example: A structural engineer must ensure a building is safe, sustainable, and compliant with regulations—lives may depend on it.

### 3. Ethics and Integrity:

Professional engineers follow a **code of ethics**, which includes honesty, accountability, fairness, and respect for others. They must avoid conflicts of interest, ensure transparency, and always prioritize safety over profits.

Example: An engineer must refuse to certify a faulty design, even under pressure from management, if it compromises safety.

## 4. License and Certification:

In many countries, engineers are required to obtain a **professional license** (e.g., PE – Professional Engineer) to legally practice and sign off on projects. This ensures they have the **competence and experience** needed to perform critical work.

### Example:

In **India**, an engineer working on a government infrastructure project—such as the design and approval of a national highway—must often be a **Chartered Engineer (CE)** registered with the **Institution of Engineers (India) [IEI]**. This certification validates their professional competence and allows them to sign technical documents, drawings, and certifications required by government agencies and regulatory bodies like the CPWD, Indian Railways, or NHAI.

In the **United States**, a civil engineer must obtain a **Professional Engineer (PE) license** to approve structural designs for public infrastructure like bridges or highways. Without this certification, they are not legally allowed to sign off on critical design documents submitted to government authorities for construction approval.



## 5. Teamwork and Communication:

Engineers often work in **multidisciplinary teams**, collaborating with architects, doctors, software developers, government bodies, and clients. Good communication and teamwork are essential to translate ideas into reality.

Example: A biomedical engineer works with surgeons to design a custom knee implant and explains the design choices in clear, simple terms.

## **6. Lifelong Learning and Innovation:**

Technology and industry needs are constantly evolving. A professional engineer must continuously upgrade skills, learn about new tools, software, materials, and methods.

Example: An electrical engineer learns about AI and IoT to implement smart energy grids in urban planning.

# Real-World Examples:

- A **civil engineer** oversees the construction of a flyover, ensuring it's strong enough to handle traffic loads and seismic forces.
- A **software engineer** develops secure banking applications used by millions of people.
- A **chemical engineer** in a pharmaceutical plant ensures that medicines are produced under safe and controlled conditions.
- A **renewable energy engineer** designs and installs solar power systems to promote sustainability.

# Comparison

| Aspect             | Engineering as Discipline                                     | Engineering as Profession                                       |
|--------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| Focus and Practice | Generation of Knowledge, theories, innovation and Research    | Practical application of knowledge to solve real-world problems |
| Setting            | Academic institutions, research labs, universities            | Industries, corporate sectors, consultancy, government agencies |
| Output             | Scientific theories, experimental results, conceptual designs | Functional products, infrastructure, processes, services        |
| Goal               | Advance the frontiers of science and technology               | Ensure safety, efficiency, and service to society               |

# Conclusion

- Engineering bridges theory and practice
- A discipline grounded in science and a profession rooted in service
- Engineers shape the future through responsible innovation